AI-Powered Cognitive Screening & Dementia Risk Analysis Platform

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Abstract*— Early identification of dementia risk is essential for timely clinical intervention; however, traditional screening approaches often rely on manual assessments and limited behavioral indicators. This study presents CogniCare, a multimodal AI-assisted cognitive screening platform integrating lifestyle profiling, cognitive and functional assessments, speech analysis, and handwriting analysis within a unified clinician-supported workflow. Collected health and behavioral data are analyzed using machine learning and standardized assessment methods to estimate dementia risk, while the results are integrated through rule-based analysis to generate an initial screening outcome. The handwriting component further introduces a browser-compatible reduced handwriting feature framework for scalable and accessible cognitive screening. Speech and handwriting modules analyze behavioral patterns associated with cognitive decline, and all outputs are consolidated into a comprehensive report available through a clinician dashboard. The platform also incorporates an interpretable clinician-assisted decision-support workflow using explainable AI and a multimodal screening pipeline deployable in low-resource environments. Guided by DSM-5-informed clinical evaluation, physicians determine whether additional neurological evaluation is required, and MRI scans can be further analyzed using deep learning-based dementia stage classification to assist clinical decision-making. The proposed platform demonstrates the potential of integrating multimodal digital assessments with clinical workflows for scalable dementia risk screening while ensuring that final diagnostic decisions remain under medical supervision. Experimental evaluation demonstrates strong performance across system modules, where the wearable-based CatBoost model achieved AUC 0.989 and 94.9% accuracy, the speech hybrid model achieved F1-score 0.82, the handwriting model achieved F1-score 0.870, and the MRI ensemble model achieved 93.75% accuracy with Macro F1-score 0.9606.

Keywords—Dementia Screening, Multimodal Health Analysis, Cognitive Assessment, Machine Learning, Speech Analysis, Handwriting Analysis, MRI Classification, Clinical Decision Support.

I. INTRODUCTION

A. Background

Dementia is a progressive neurological disorder that affects cognitive functions such as memory, language, reasoning, and the ability to perform everyday activities [1]. It is one of the leading causes of disability and dependency among older adults worldwide. The number of individuals living with dementia is rapidly increasing due to ageing populations, placing significant pressure on healthcare systems.

Early detection of cognitive decline is critical because timely interventions can slow disease progression, improve patient management, and enhance quality of life. [2] Traditional dementia diagnosis often relies on clinical assessments, cognitive screening tests, and neuroimaging techniques conducted in specialized healthcare settings. [3] Common screening tools such as cognitive assessments and functional ability evaluations provide valuable insights into memory, language, and executive functioning.

However, these assessments are typically conducted periodically in clinical environments and require trained medical professionals. As a result, they may not capture subtle behavioral and physiological changes that occur in the early stages of cognitive decline. Recent advances in artificial intelligence, wearable health technologies, and digital behavioral analysis offer new opportunities for developing automated systems capable of supporting early dementia screening using multiple data sources.

Behavioral indicators such as speech patterns, handwriting dynamics, and daily activity levels [4], together with medical imaging data such as brain MRI scans, can provide important signals related to cognitive health [16]–[18]. By integrating these heterogeneous data sources into a unified analytical framework, intelligent systems can assist clinicians in identifying early signs of cognitive impairment more effectively.

B. Problem Statement

Despite progress in dementia research, current screening approaches still face several limitations. Traditional cognitive assessments are often manual and require face-to-face clinical evaluations, making them time-consuming and difficult to scale for large populations. In addition, many existing methods rely on single data modalities such as neuroimaging, cognitive test scores, or behavioral analysis, which limits their ability to capture the multidimensional nature of cognitive decline. Although lifestyle and physiological factors such as physical activity, sleep patterns, and mental health are important contributors to dementia risk, these indicators are rarely integrated into current screening systems. As a result, the lack of automated multimodal platforms that combine

behavioral, physiological, and neuroimaging data restricts the development of scalable and accessible dementia screening solutions.

C. Research Gap

Existing dementia detection research mainly focuses on isolated approaches such as MRI analysis, speech analysis, handwriting assessment, or wearable monitoring independently. Most existing systems lack effective cross-modal fusion that combines behavioral, physiological, cognitive, and neuroimaging indicators within a unified framework.

In addition, many current approaches provide limited explainability and are not designed for real-time or low-resource deployment. Several systems also depend on specialized hardware and are not compatible with lightweight browser-based environments. These limitations highlight the need for an interpretable, browser-compatible, and multimodal dementia screening platform. Therefore, this research proposes CogniCare, a clinician-assisted system that integrates wearable monitoring, cognitive assessment, handwriting analysis, speech analysis, and MRI-based evaluation within a unified decision-support workflow.

D. Objective

The objective of this research is to design and develop CogniCare, an AI-powered multimodal cognitive screening and dementia risk analysis platform that integrates handwriting behavior analysis, wearable health and lifestyle monitoring, speech-based cognitive assessment, and MRI-based neuroimaging classification.

The research further aims to develop a browser-compatible reduced handwriting feature framework capable of supporting scalable and accessible cognitive screening across low-resource and web-based environments. In addition, the platform is designed to provide an interpretable clinician-assisted decision-support workflow using explainable AI (XAI) techniques to improve transparency and clinical usability during dementia risk evaluation.

By integrating these diverse data sources into a unified analytical framework, the proposed system aims to support early dementia detection, provide clinically interpretable decision-support insights for physicians, and enable scalable multimodal cognitive health monitoring in both clinical and home-based environments while ensuring that final diagnostic decisions remain under medical supervision.

II. Related Work

Digital cognitive screening tools improve the reliability of traditional paper-based assessments. Newman et al. [7] introduced ACE mobile, a tablet-based version of ACE-III that reduces scoring errors through automated scoring and standardized administration. This demonstrates the benefits of computerized cognitive assessments in improving diagnostic consistency.

A. Smart Health Monitoring & Dementia Screening Risk Analysis

Wearable devices enable continuous monitoring of physiological signals, activity, and sleep behavior associated with dementia risk. Iaboni et al. [8] used wearable sensors and machine learning to detect behavioral symptoms of dementia, while Chan et al. [9] found that walking speed, physical activity, and sleep patterns are significantly associated with dementia risk.

B. Handwriting and Drawing-Based Assessment

Handwriting and drawing tasks have been explored as digital biomarkers for cognitive decline. Impedovo et al. [10] demonstrated that handwriting dynamics can reflect cognitive and motor impairments. Xu et al. [11] and De Stefano et al. [12] further showed that temporal and kinematic handwriting features can support machine learning–based detection of neurodegenerative diseases.

C. Dementia screening through guided speech-task analysis

Speech and language changes are recognized as early indicators of cognitive decline. Lopes da Cunha et al. [13] applied automated speech analysis to identify linguistic patterns associated with Alzheimer's disease. Luz et al. [14] introduced the ADReSS dataset for dementia detection from speech recordings, while Yeung et al. [15] confirmed strong correlations between automated speech features and clinician rated impairments.

D. Dementia stage classification using MRIs

Structural MRI is widely used for automated dementia detection. Lu et al. [16] developed a large-scale deep learning classifier using multiple MRI datasets. Popuri et al. [17] and Liu et al. [18] demonstrated that machine learning and deep learning models can effectively distinguish cognitively normal individuals from patients with mild cognitive impairment and Alzheimer's disease. Inspired by these studies, the proposed system applies an ensemble of CNN models for MRI-based dementia stage classification with explainable AI techniques.

III. Methodology

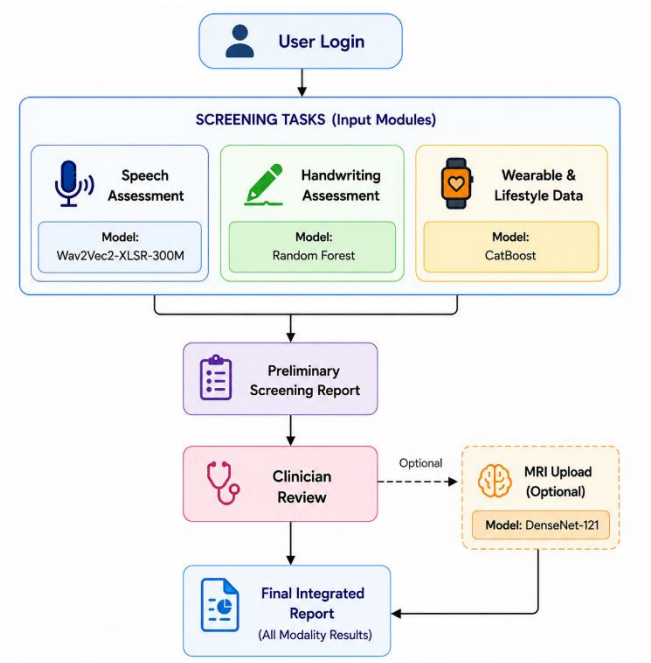


Fig. *1*. System Overview

"Fig. 1" illustrates the overall workflow of the proposed CogniCare platform. After user authentication, the system performs speech assessment, handwriting analysis, and wearable/lifestyle-based cognitive screening. Each module independently generates modality-specific cognitive

risk indicators, which are presented through a unified clinician dashboard. Based on these screening results, clinicians determine whether additional neurological evaluation is required. If necessary, structural MRI scans can be uploaded for dementia stage classification using deep learning models. The platform functions as a clinician-supported decision-support system where physicians collectively interpret screening and MRI findings to support dementia risk evaluation and clinical decision-making.

A. Smart Health Monitoring & Dementia Screening Risk Analysis

The Smart Health Monitoring module collects multimodal health, behavioural, and assessment data through the CogniCare application. Demographic and lifestyle information includes age, gender, education level, family history of dementia, smoking status, alcohol consumption, diet type, chronic conditions, diabetes status, sleep behaviour, and PHQ-2 depression indicators. Physiological indicators such as heart rate, step count, and physical activity are obtained through wearable device integration using Google Fit. Cognitive performance is assessed using ACE-III tasks, while functional ability is evaluated using the Lawton Instrumental Activities of Daily Living (IADL) scale.

The collected data undergo preprocessing including data cleaning, missing value handling, and normalization, while categorical variables are processed using CatBoost's native categorical handling. Feature engineering generates additional indicators such as heart-rate-to-age ratios, aggregate statistics, and missing-value indicators to capture physiological and behavioural patterns. A CatBoost gradient boosting classifier is then used to estimate dementia risk from lifestyle and physiological indicators, with hyperparameter optimization performed using Optuna and evaluation conducted using Repeated Stratified K-Fold cross-validation. The model outputs probability scores mapped into three categories: Low Risk, At Risk, and High Risk, achieving an AUC of 0.9892 and an accuracy of 94.9%.

Cognitive assessment produces ACE-III scores across attention, memory, fluency, language, and visuospatial domains, classified as Normal (≥88), At Risk (82–87), or Declining (<82). Functional ability is evaluated using the Lawton IADL scale, producing states categorized as Independent, Partially Dependent, or Dependent. To maintain interpretability, the final screening result is determined through a rule-based fusion mechanism combining wearable risk predictions, cognitive assessment results, and functional dependency states. These indicators are mapped into dementia risk categories including Normal, Very Mild, Mild, Moderate, and Severe, and the results are presented through a clinician dashboard to support medical review and further evaluation.

B. Handwriting-Based Dementia Screening

The handwriting module analyzes digital writing behavior to identify motor and cognitive patterns associated with early dementia, as changes in handwriting dynamics may reflect impairments in memory, motor coordination, and executive control. The module was developed using the DARWIN handwriting dataset [19], which contains digital handwriting recordings from individuals with and without cognitive impairment represented as sequences of spatial coordinates and timestamps captured during writing and drawing tasks.

Within the CogniCare platform, handwriting data are collected through a browser-based drawing interface where users perform predefined tasks while the system records x–y coordinates and timestamps of pen movements. From these trajectories, dynamic behavioral features such as stroke speed, acceleration, movement duration, inter-stroke timing, and displacement characteristics are extracted to characterize handwriting rhythm and performance. Prior studies have demonstrated that handwriting dynamics and temporal motor features can provide useful indicators of cognitive and neurological impairment [10], [12].

The feature reduction process was performed to improve browser compatibility, deployment efficiency, and computational performance while preserving clinically meaningful handwriting characteristics associated with cognitive decline. The original Full Model contained 452 features extracted from 25 handwriting and drawing tasks available in the DARWIN dataset [19]. However, many of these tasks and sensor-dependent variables were not suitable for browser-based deployment environments.

To improve usability and compatibility within the CogniCare platform, the task set was reduced to eight browser-friendly handwriting tasks (Tasks 2, 3, 4, 5, 8, 9, 21, and 24). Task selection was guided by findings from prior handwriting-analysis studies [10], [12] together with browser compatibility, usability, and deployment considerations to ensure reliable handwriting capture in web-based environments while preserving cognitively meaningful writing patterns. The selected tasks included both simple motor tasks and cognitively demanding writing and drawing activities to preserve behavioral diversity.

In addition, noisy and device-dependent variables including in-air movement, pressure, tilt, and azimuth features were removed because such signals are not consistently supported across browser-based devices. This process produced the Browser Model consisting of 96 browser-compatible handwriting features.

Subsequently, SHAP-based feature importance analysis and iterative model evaluation were applied to identify the most discriminative behavioral indicators. Using this process, the feature space was further reduced to 40 features in the Reduced Model and 24 features in the Ultra Reduced Model while preserving discriminative behavioral information related to cognitive impairment.

The Reduced Model primarily retained clinically meaningful handwriting dynamics features including mean writing speed, mean acceleration, mean jerk, total writing time, and displacement index across the selected handwriting tasks. These features were selected because prior research has shown that movement smoothness, writing speed, motor stability, and execution time are strongly associated with cognitive and neurological impairment [10], [12].

After preprocessing, missing values were handled using median imputation and features were standardized using z-score normalization. Machine learning models including Random Forest and XGBoost were trained to capture non-linear relationships between handwriting dynamics and cognitive status. SHAP (SHapley Additive Explanations) was applied to analyze feature importance and identify handwriting characteristics contributing to cognitive impairment predictions.

C. Dementia screening through guided speech-task analysis

The speech module is designed to identify early cognitive risk through structured guided speech elicitation, as speech production reflects memory retrieval, lexical access, executive control, and motor coordination. Speech data are collected through three guided tasks: self-introduction, morning recall, and picture description selected to capture verbal fluency, episodic memory, and narrative coherence. Audio responses are recorded through a browser-based interface and stored with session metadata.

The primary speech classification pipeline is based on a fine-tuned Wav2Vec2-XLSR-300M architecture, a transformer-based multilingual speech representation model trained on large-scale speech data. Speech recordings are resampled to 16 kHz and processed to extract high-level acoustic embeddings together with speech characteristics such as duration, tempo, zero-crossing rate (ZCR), and root mean square (RMS) energy to characterize fluency and speech rhythm.

In addition to the acoustic pipeline, transcript-based linguistic analysis and hybrid fusion approaches were also evaluated. Recordings were automatically transcribed, and linguistic features including word count, sentence count, speech rate, and lexical variation were extracted to capture memory-related language organization and narrative structure. Experimental evaluation showed that the standalone Wav2Vec2-XLSR-300M model achieved the best overall performance and was therefore selected as the final speech classification model within the CogniCare platform.

The module generates speech-based risk scores and clinician-facing explanation outputs related to fluency, memory-related language, and narrative coherence to support interpretable dementia screening. The system functions strictly as a decision-support tool rather than a standalone diagnostic system.

D. MRI-Based Dementia Stage Classification

The MRI-based dementia classification module analyses structural brain MRI scans to predict dementia severity and support clinical decision-making. The system performs four-class classification consisting of Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented stages.

The models were trained using T1-weighted structural MRI scans obtained from publicly available datasets such as OASIS, where images are labelled using clinically verified diagnoses or Clinical Dementia Rating (CDR) scores. The MRI volumes were pre-processed through skull stripping, intensity normalization, and spatial alignment. Each 3D scan was converted into 2D slices suitable for convolutional neural networks, resized to 224 × 224 pixels, and normalized using ImageNet statistics.

Three convolutional neural network architectures, DenseNet-121, EfficientNet-B0, and ResNet-50, were trained and evaluated for dementia stage classification. The original classification layers were replaced with task-specific fully connected layers and a four-class Softmax output layer. Models were trained using cross-entropy loss and the Adam optimizer, with data augmentation techniques such as random rotations and horizontal flips applied to improve generalization. Early stopping based on validation performance was used to reduce overfitting.

A weighted ensemble approach combining the outputs of the three CNN models was also evaluated to improve prediction robustness. However, DenseNet-121 achieved the highest individual performance in terms of accuracy and Macro F1-score and was therefore selected as the final model for dementia stage prediction within the CogniCare platform.

To enhance interpretability, Grad-CAM++ was applied to generate heatmaps highlighting brain regions influencing model predictions. Within the CogniCare platform, MRI classification is used as an optional advanced assessment when imaging data is available. The predicted stage and confidence scores are integrated into the clinician dashboard alongside results from lifestyle monitoring, cognitive assessments, handwriting analysis, and speech evaluation to support comprehensive dementia risk screening.

IV. Results & Evaluation

The effectiveness of the proposed CogniCare system was evaluated using standard machine learning metrics including Accuracy, Precision, Recall, F1-score, and Area Under the ROC Curve (AUC). These metrics measure the model's ability to correctly identify dementia-related patterns.

The evaluation was conducted component-wise, analyzing the performance of the wearable-based dementia risk prediction model, handwriting analysis model, Speech analysis model, and MRI analysis model. The results demonstrate that the selected machine learning approaches provide reliable performance for early dementia risk screening.

A. Smart Health Monitoring & Dementia Screening Risk Analysis

The Smart Health Monitoring module uses a CatBoost gradient boosting classifier to estimate dementia risk based on physiological and lifestyle indicators. The model was trained in the processed clinical dataset using stratified cross validation.

The results demonstrate strong predictive performance, achieving an accuracy of 94.9% and an AUC score of 0.989, indicating effective discrimination between dementia and non-dementia cases. The confusion matrix further confirms that the model correctly classifies most cases, highlighting the suitability of CatBoost for heterogeneous healthcare data. The performance metrics of the CatBoost dementia risk model are presented in Table I.

TABLE I. Performance Metrics Of Catboost

Metric	Value
Accuracy	0.949
Precision	0.95
Recall	0.95
F1 Score	0.95
AUC	0.989

B. Handwriting-Based Dementia Screening

The handwriting analysis module evaluates writing dynamics to detect cognitive impairment patterns. Multiple feature configurations and machine learning algorithms were evaluated to determine the most effective model for handwriting-based dementia screening.

Two classification algorithms, Random Forest and XGBoost, were evaluated using different feature sets, including full, browser-based, and reduced configurations. As shown in Table II, the 40-feature reduced model achieved a competitive F1-score of 0.870 while significantly reducing feature dimensionality. Therefore, this configuration was selected as the final handwriting classification model due to its balance between performance and computational efficiency.

TABLE II. HANDWRITING MODEL PERFORMANCE

Model Version	Features	Models Trained	Best Model	Best F1	AUC
Full Model	452	Random Forest + XGBoost	XGBoost	0.886	0.950
Browser Model	96	Random Forest + XGBoost	XGBoost	0.869	0.942
Reduced Model	**40**	Random Forest + XGBoost	**Random Forest**	**0.870**	**0.932**
Ultra Reduced	24	Random Forest + XGBoost	Random Forest	0.838	0.934

Although the Full Model achieved the highest overall performance due to the availability of richer stylus and sensor-derived signals, the Reduced Model achieved competitive performance while significantly lowering feature dimensionality. The Reduced Model also slightly outperformed the Browser Model, suggesting that removing redundant and noisy variables improved model generalization and reduced overfitting. Therefore, the 40-feature Reduced Model provided a better balance between predictive performance, browser compatibility, and deployment efficiency within the CogniCare platform.

C. Dementia screening through guided speech-task analysis

The speech-based dementia screening module analyzes guided speech responses to detect cognitive impairment. Three models were explored: a Wav2Vec2-XLSR-300M deep learning model for raw audio classification, an acoustic feature model using Random Forest, and a transcript-based model using XGBoost. In addition, a hybrid fusion model combining Wav2Vec2-XLSR-300M and transcript-based linguistic features was evaluated to determine whether integrating acoustic and semantic speech characteristics could improve classification performance. The performance comparison of the speech-based classification models is presented in Table III.

As shown in Table III, the Wav2Vec2-XLSR-300M model achieved the best overall performance, while the hybrid fusion model produced comparable results. Although the fusion approach provided a more comprehensive representation of speech characteristics, it did not significantly outperform the standalone Wav2Vec2-XLSR-300M model. Therefore, Wav2Vec2-XLSR-300M was selected as the final speech classification model within the CogniCare platform due to its strong predictive performance and simpler architecture.

TABLE III. PERFORMANCE METRICS OF SPEECH

Model	Accuracy	Precision	Recall	F1 Score	AUC
Wav2Vec2-XLSR-300M	**0.79**	**0.80**	**0.77**	**0.78**	**0.85**
Acoustic Model (Random Forest)	0.73	0.74	0.70	0.71	0.79
Transcript Model (XGBoost)	0.76	0.77	0.74	0.75	0.84
Hybrid Fusion Model	0.79	0.81	0.79	0.78	0.85

Although the hybrid fusion model achieved performance comparable to the standalone Wav2Vec2-XLSR-300M model, the integration of acoustic and transcript-based features provided a more comprehensive representation of speech characteristics related to cognitive decline. The fusion approach combines both semantic and acoustic information, improving the robustness and interpretability of speech-based dementia screening.

D. MRI-Based Dementia Stage Classification

The MRI analysis component classifies dementia stages from brain MRI images using deep learning models. Several convolutional neural network (CNN) architectures were evaluated to identify the most reliable model for dementia stage prediction.

Among the tested models, **DenseNet-121, EfficientNet-B0, and ResNet-50** showed the best performance. An additional weighted **ensemble** approach combining the outputs of these three networks was also evaluated to analyze whether multi-model aggregation could improve prediction robustness.

As shown in Table IV, DenseNet-121 achieved the highest overall performance across Accuracy, Macro F1-score, Macro Precision, and Macro Recall metrics. Although the ensemble model produced competitive results, it did not outperform DenseNet-121. Therefore, DenseNet-121 was selected as the final MRI classification model within the CogniCare platform due to its superior predictive performance and model efficiency.

TABLE IV. PERFORMANCE OF SELECTED MRI CLASSIFICATION MODELS

Model	Accuracy	Macro F1	Macro Precision	Macro Recall
DenseNet-121	**0.9484**	**0.9657**	**0.9633**	**0.9684**
EfficientNet-B0	0.8562	0.9067	0.9085	0.9171
ResNet-50	0.8406	0.8921	0.8957	0.8889

Ensemble Model	0.9375	0.9606	0.9591	0.9636

V. Discussion and Conclusion

This study introduced CogniCare, a multimodal AI-assisted dementia screening platform integrating wearable health monitoring, cognitive assessments, handwriting analysis, speech-based evaluation, and MRI-based stage classification within a unified framework.

Experimental results demonstrated promising performance across all system components. The Smart Health Monitoring module achieved an AUC of 0.989 and 94.9% accuracy using a CatBoost model for dementia risk estimation. The handwriting analysis module showed that writing dynamics can indicate cognitive impairment, where a reduced-feature Random Forest model achieved an F1-score of 0.870. The speech analysis module demonstrated the effectiveness of guided speech assessment for cognitive screening, where the Wav2Vec2-XLSR-300M model achieved the best performance with an F1-score of 0.78 and an AUC of 0.85. In MRI-based analysis, DenseNet-121, EfficientNet-B0, ResNet-50, and a weighted ensemble approach were evaluated. DenseNet-121 achieved the best overall performance and was selected as the final MRI classification model, achieving 94.84% accuracy and a Macro F1-score of 0.9657.

By combining outputs from multiple modalities, the proposed system improves screening robustness and provides interpretable reports to support clinicians in early dementia risk assessment. The platform further contributes an interpretable clinician-assisted decision-support workflow through explainable AI techniques such as SHAP and Grad-CAM++. In addition, the multimodal screening pipeline is designed for low-resource and browser-based environments using reduced-feature handwriting models and lightweight web-based assessment workflows. CogniCare is designed as a decision-support screening tool, ensuring that final diagnostic decisions remain under medical supervision.

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